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LinkedIn | GitHub | Portfolio

SUMMARY

Software engineer with full-stack experience at Light & Wonder and Liminal Custody, plus co-founder engineering work at Medha Innovations. Builds backend services with Go, TypeScript, PostgreSQL and Redis, alongside Python/FastAPI applied-AI portfolio projects. Focuses on transactional correctness, authorization, structured model outputs, evaluation and human review. Recent work includes an approval-gated incident-triage workflow, evidence-linked document reconciliation and an isolated single-host observability deployment. Brings experience in SQL-backed workflows, API security, debugging and automated testing, with project evidence that separates synthetic benchmarks, prototypes and deployed behavior.

TECHNICAL SKILLS

- Languages and Frameworks: Python, TypeScript, Go, SQL; FastAPI, Pydantic, asyncio, Node.js, REST APIs, Server-Sent Events (SSE), WebSockets, React and Angular.
- LLM and Workflows: Gemini, Groq and OpenAI-compatible APIs; prompt design, structured outputs, JSON Schema, tool calling, LangGraph, Model Context Protocol (MCP), checkpointed workflows, content-bound approvals and bounded execution.
- Retrieval and Search: PostgreSQL, SQLite FTS5, BM25, embeddings, hybrid retrieval, reciprocal rank fusion, ONNX Runtime, cross-encoder reranking and quantization.
- Document AI: Docling, OCR, vision-language models, Pydantic-validated extraction, evidence bounding boxes, decimal-exact reconciliation and human-review routing.
- Testing and Observability: pytest, Go race tests, held-out and synthetic evaluations, F1, Recall@k, latency/cost tracking, OpenTelemetry, Prometheus, Grafana and Alertmanager.
- Data and Delivery: PostgreSQL, PostGIS, Redis, SQLite, SQL Server, Docker, GitHub Actions and GitLab CI/CD; Kubernetes (k3s) and Argo CD deployment configuration.

PROJECTS

ReconcileAI: Document AI Reconciliation | Python, FastAPI, Docling, Gemini, Pydantic | GitHub

- Built a local invoice/PO workflow combining Docling parsing, Pydantic-validated model extraction and decimal-exact discrepancy checks, with source-evidence links and human review.
- Compared extraction on 20 held-out public CORD receipts: Gemini vision reached 76.0% field F1 versus 66.0% for a Qwen VLM and 56.9% for OCR plus text LLM; not invoice/PO accuracy.
- Verified deterministic controls on 50 labelled synthetic invoice/PO pairs; separately measured live model behavior, including review burden and row-scoring limitations. Streams progress over SSE.

TraceWard: Incident-Triage Agent | Python, FastAPI, LangGraph, MCP, OpenTelemetry | GitHub

- Built a checkpointed, predefined LangGraph workflow for log parsing, similar-incident retrieval, root-cause suggestions and fix drafting, using MCP tools over stdio and OpenTelemetry tracing.
- Persists a local ticket draft only after authenticated, content-bound approval; enforces tenant scope, signed approval, idempotency and execution budgets. Does not file external issues or execute fixes.
- Evaluated 50 synthetic known-family holdouts against a 200-incident corpus: 50/50 root-cause top-1; 20 approval-bypass probes produced no unauthorized drafts. Not general incident accuracy or security.

Homelab SRE: Observability | Go, Docker, Prometheus, Grafana | GitHub

- Deployed an isolated six-service observability stack on a shared Linux host: loopback-only access, file-backed secrets, authenticated alert receiver, disabled fault controls and resource limits.
- Added CI for Go race tests, alert lifecycle, restart recovery and cold-backup/file-restore checks; coverage-gated SLOs suppress warm-up claims. Single-host deployment, not HA or independent paging.

PROFESSIONAL EXPERIENCE

Founding Engineer / Co-founder — Mar 2026 – Sep 2026

Medha Innovations | Self-funded, two-founder booking and messaging venture.

- Co-founded a booking and messaging venture and designed its Go modular backend across 21 domains. Built REST APIs for profiles, practitioner discovery, availability and bookings, with PostgreSQL persistence and explicit service boundaries.
- Designed booking workflows around PostgreSQL transactions and row locking to coordinate competing requests, with PostGIS for location-based practitioner discovery. Organized the corresponding data models and API boundaries within the modular backend.
- Platform design combined RS256 JWTs and RBAC for role-aware access with Redis Pub/Sub-backed WebSockets for messaging. Integrated authentication, booking and real-time delivery concerns within explicit modular service boundaries.
- Collaborated with a co-founder on backend integration and deployment planning for the self-funded product, covering the Go API, PostgreSQL and Redis dependencies. Project configuration includes Docker and Kubernetes/Argo CD deployment paths.

Software Engineer, Full Stack — Nov 2025 – Mar 2026

Liminal Custody (First Answer India Services Pvt. Ltd.) | Digital asset custody.

- Built a transaction-policy engine for three controls (TRM Labs address-risk screening, FATF Travel Rule checks, transfer velocity limits) as TypeScript APIs with Redis-cached, short-circuit rule evaluation.
- Implemented tenant-level RBAC, quorum approvals, Auth0 authentication and step-up two-factor authentication (2FA) with signature validation, plus TypeORM transactions that prevent partial updates.

Senior Associate Software Engineer (final role) — Aug 2023 – Jul 2025

Light & Wonder | Full-time: Aug 2023–Jul 2025; separate internship: Mar–Jul 2023.

- Built an internal LLM-assisted debugging tool (Gemini API, React UI), deployed on Microsoft Azure with the DevOps team and used by internal teams, that matched crash reports and telemetry to similar historical hardware incidents and their fixes.
- Selected for the slot-machine modernization team; built an in-game LLM chat assistant (OpenAI API, Angular) that lets guests order food and beverages and choose music from their seat; demoed to management and approved for lab testing.
- Fixed a memory leak that made slot machines hang or restart after about one hour: reproduced it on hardware, corrected component cleanup and resource reuse, and verified sustained operation.
- Built SQL Server-backed APIs and screens for casino cash operations; automated audit capture with triggers and stored procedures; wrote Cypress end-to-end tests.
- Worked on CONNEXUS, the slot-machine tracking and data management system that monitors machine status and usage and collects data on a central server; contributed to performance profiling and monitoring.

EDUCATION

Siddaganga Institute of Technology (SIT), Tumakuru — 2019 – 2023

Bachelor of Engineering (B.E.), Computer Science and Engineering — CGPA: 8.65/10

PUBLICATIONS AND CERTIFICATIONS

"Data Visualisation of Time Tradable Assets Using Machine Learning," IEEE NMITCON — 2023

DOI 10.1109/NMITCON58196.2023.10275962 | ResearchGate

Model Context Protocol (MCP): Introduction and Advanced Topics | Anthropic

Docker Foundations Professional Certificate | LinkedIn Learning